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SEXUAL ISOLATION IN *DROSOPHILA*

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Isolation is an indispensable factor in the origin and maintenance of race and species. Sexual or ethological isolation, a barrier to random mating due to behavioural incompatibilities, constitutes the largest and most important class of reproductive isolation in animals. The phenomenon of ethological isolation is widespread in the genus *Drosophila*, commonly known as fruit fly. The pattern of mating and the degree of sexual isolation have often been used to elucidate the direction of evolution among closely related species.

ISOLATION may be defined as the prevention of gene flow between Mendelian populations. Even Lamarck and Darwin recognized the importance of isolation in evolution. It has been suggested that the formation of races and species without isolation is impossible. It was Th. Dobzhansky of Columbia University, New York, USA who coined the term "isolating mechanisms" as a generic name for all such agents which prevent gene exchange between populations. If reproductive isolating mechanisms do not operate, the genetic differences which are pre-requisite for speciation and acquired through natural selection and genetic drift, will be swamped and speciation is not possible. Species of sexually reproducing organisms are genetically closed systems, because there is no gene exchange between them due to genetically conditioned barriers. According to Ernst Mayr of Harvard University, USA, the isolating mechanisms are the most important attributes a species has. Gene exchange between *allopatric populations* (those populations which are inhabiting different geographical areas) is prevented because individuals do not get a chance to mate.

This constitutes geographical isolation. Such populations may or may not show reproductive isolation if brought together under laboratory conditions. On the other hand, *sympatric populations* (inhabiting the same geographical area) can be maintained as distinct units only if interbreeding is prevented by barriers which constitute reproductive isolating mechanisms.

There are different factors which operate as reproductive isolating mechanisms between Mendelian populations. One or more mechanisms in combination serve to prevent gene exchange between separate Mendelian populations. Various authors have discussed and classified the kinds of reproductive isolating mechanisms.

Kinds of reproductive isolating mechanisms :

- (i) *Seasonal isolation*—The individuals are sexually functional at different times.
- (ii) *Ecological isolation*—Individuals of populations are restricted to different habitats of the same area by biological needs.

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- (iii) *Mechanical isolation*—There is physical incompatibility between the sexual parts.
- (iv) *Sexual or ethological isolation*—Lack of mutual attraction between males and females of different species due to behavioural incompatibilities.
- (v) *Gametic isolation*—Sperm of one species are not attracted to the eggs or poorly viable in the genital tracts of another species.
- (vi) *Hybrid inviability*—F₁ hybrid zygotes are produced, but they are inviable.
- (vii) *Hybrid sterility*—F₁ hybrids produced are sterile individuals and unable to produce normal functional gametes.

All these isolating mechanisms either alone or in combination operate to prevent interbreeding between Mendelian populations which are sympatric. All mechanisms have a genetic basis. Sexual or ethological isolation which may be defined as the lack of mutual attraction between males and females of different species constitutes the largest and most important class of reproductive isolating mechanisms in animals. The individuals respond in such a manner to the stimuli they receive from the individuals of the opposite sex belonging to other species, that mating does not occur. Thus, the word ethological refers to behavioural pattern. Ethological isolation becomes a barrier to mating due to incompatibilities in behaviour. Males of one species have specific courtship pattern to which only females of the same species are receptive. The specific reactions of males and females towards each other is known as *species recognition*. In most cases, it is the males that actively search for a mate. *Drosophila*, a dipteran insect commonly known as fruit fly, has been found to be a very useful material for behavioural studies. In 1915, A. H. Sturtevant of Columbia University, USA, was the first to study courtship pattern of *D. melanogaster*. Since then, mating behaviour of numerous *Drosophila* species has been extensively studied by several investigators. Mating behaviour of *Drosophila* consists of species-specific actions which are accompanied by orientation movements. Such actions referred to as courtship displays are made up of several

signals which are performed sequentially. Mating occurs only in the female responds by performing acceptance signals. The courtship is initiated by the male which performs a fixed action pattern directed towards the female. The response of potential mate produces informations through which individuals discriminate between conspecific and nonconspecific and the physiological readiness of females to engage in copulation is also known. If the individual is a nonconspecific female, courtship is quickly terminated. Mating takes place only if the potential male is a conspecific receptive female. The forelegs, wings and mouth parts of males are used as signalling organs in most of the species. Male display presents a great diversity among different species. However, the female signals are less in number and diversity than males and are performed by wings, legs, genitalia and abdomen. Females produce signals in response to male courtship displays and these signals are of two types: acceptance response and rejection response. It has been observed that females may walk and even feed also during copulation. But males are inert during copulation. The duration of copulation varies considerably among different species. It ranges from 5 sec in *D. enigma* to 62 min in *D. acanthoptera*.

The phenomenon of sexual or ethological isolation is widespread in *Drosophila* due to which gene exchange between different species is prevented. A large number of investigations have been carried out in different species of *Drosophila* by several workers who have demonstrated that complete sexual isolation exists between some species but is incomplete between others. It is known that it is the females which exercise discrimination during courtship. Mating isolation is based on production and reception of various stimuli by the sex partners. These behavioural devices have classified on the basis of the main sense organs involved.

Different types of stimuli involved in mating behaviour

Ethological isolation is based on the production and reception of stimuli by male and

female. On the basis of the main sense organs involved, these stimuli have been classified into different types: visual, auditory and chemical. A male *Drosophila* provides the female a set of stimuli during courtship. Based on these stimuli, the female is able to discriminate between its own and alien males and is stimulated for copulation.

(1) *Visual stimuli*: The role of visual stimuli in mating has been detected in different species of *Drosophila*. All species court during the period of day light and some also court in darkness. However, no species restricts its courtship only during darkness. It has been shown that the presence or absence of light has a profound effect on mating of *Drosophila*. Various species exhibit varying degrees of light dependency in their mating behaviour. They have been classified into three categories by J. Grossfield of City College of New York, USA in 1971:

(a) *Light independent*: the species mate effectively in darkness also; 2-light dependent—the species are unable to mate in darkness and 3-facultative species—which are depressed by darkness but can achieve some copulations. In light-dependent species (*D. subobscura*) important components of courtship of males appear to be visual. The role of visual cues in *Drosophila* mating has also been detected while studying the mating behaviour of certain eye pigment deficient mutants. For example, white-eye mutants of different species are at a disadvantage in mating in light conditions. Thus, vision plays an important role in *Drosophila* courtship. The inhibition of mating of white-eye mutants in light in different species of *Drosophila* has been explained by suggesting different hypotheses. According to H. T. Spieth of University of California, Davis, USA, eye-pigment deficient mutants of *D. melanogaster* exhibit a deficit of optomotor response. A. W. Ewing and A. Manning of University of Edinburgh, UK, suggested that the mutation in eye pigmentation have some pleiotropic effects which affect the central nervous system of the animals which, in turn, affects the male

sexual behaviour. On the other hand, J. Grossfield and his associates of the City College of New York, are of the view that mating deficit of white-eye males of *D. pseudoobscura* in light is due to a neurobehavioural disruption produced by faulty visual input. Further more, orientation of flies can not occur in darkness in light-independent and facultative species but mating may occur due to accidental encounters or as a result of searching movements by the males. The light-dependent species remain immobile in darkness. Hawaiian *Drosophila* species remain immobile during darkness in field and *D. subobscura* females perform dance which must be observed visually by males. In certain species of *Drosophila* (e.g. picture-winged Hawaiian *Drosophila*, *D. suzukii*, *D. biarmipes*), males possess dark black patch on their wings which serve as a visual stimulus to the female during courtship. Removal of the black shade in wings by amputation reduces mating success of males. In *D. biarmipes*, both types of males, with or without patch, have been found. Males with black patch on their wings have greater mating success than those without patch. Thus visual stimuli play an important role in mating in certain species of *Drosophila*.

(b) *Auditory stimuli*: Songs, calls and other acoustic signals have precision and specificity; they become more suitable isolating mechanisms than visual stimuli. *Drosophila* females may not become receptive for mating until stimulated by appropriate wing vibration of conspecific males. Males of most *Drosophila* species make wing vibrations which cause the production of sound termed as male courtship songs. These songs have been studied by different workers in different species. The sex songs of males consist of two components: the sine song and the pulse song. The pulse song consists of repeated sound pulses separated by time or interpulse intervals. With respect to pulse songs, there is variation among the different species and also within the species in the number of cycles, pulse-repetition rate and the duration of interpulse intervals. The antennae

are the main organs to receive auditory signals in both sexes. Flies of both sexes which are devoid of wings and antennae are mute and deaf. The male sex songs are species-specific. The interpulse interval of male pulse songs is most important and is discriminated by females. Thus, it forms the basis for ethological isolation among different species. When incipient isolation occurs among different strains of the same species, variation in male pulse songs has been noted in some cases. Thus, male sex organs appear to function in mate discrimination by the female and in her final acceptance of copulation.

(c) *Chemical stimuli*: Chemical stimuli which act at a distance (olfactory) or on contact often serve as isolating mechanisms. It has been shown that olfactory stimuli in the form of pheromones are involved in mating behaviour of *Drosophila*. In both sexes, the receptors for olfactory stimuli are present on the third antennal segment. During courtship, the physical contacts between the male and female also results in the transmission of chemostimuli. The male usually taps the female by its forelegs. Chemoreceptors are present on the foretarsi of males and females have chemoreceptors on their body surface. Chemostimuli are exchanged during tapping of the female by the male. The role of chemostimuli in mating has also been shown in Hawaiian *Drosophila* species in which males of most species engage in lek behaviour (courtship occurs in a limited area away from feeding site). It has been reported that volatile pheromones are produced by both sexes which are essential for the initiation of courtship. The role of chemostimuli in mating isolation has also been demonstrated by amputation of antennae which reduces the ability of *D. pseudoobscura* and *D. persimilis* females to discriminate between their own and alien males.

Thus, there is ample evidence to show that during courtship exchange of various stimuli between the male and female takes place and mating occurs only when both sex partners reach a state of physiological readiness. Furthermore,

various stimuli play important role in mating isolation.

Mode of mating preference and direction of evolution

Sexual isolation has been observed in many species of *Drosophila*. However, the mode and the degree of isolation varies in different cases. In many cases, complete sexual isolation has been observed. Numerous cases of incomplete isolation have been reported among closely related species. Moreover, the mode of mating preference may also vary. The species which are incompletely isolated may show symmetrical (in both directions or both-sided) or asymmetrical (in one direction or one-sided) sexual isolation. Some times geographic populations or strains of the same species may show incipient isolation as a result of genetic divergence caused due to natural selection and genetic drift. The degree and pattern of mating preference have often been used to know the phylogenetic relationships among closely related species. Recently, two opposing models have been proposed to elucidate the evolutionary sequence of species on the basis of the mode of mating preference between closely related species.

Sexual isolation is caused due to preferential (non-random or assortative) mating between males and females of the same species. This is known as homogamic mating. In the case of incomplete isolation, mating may take place between males and females of different species which is referred to as heterogamic mating. Sexual isolation (or the pattern of mating) may be symmetrical or asymmetrical. The occurrence of asymmetrical isolation has been reported in many *Drosophila* species groups.

Table 1 shows that homogamic matings are more frequent than heterogamic ones and isolation indices remain high (close to 1) in both directions. Thus, there is evidence for preference for homogamic matings in both directions. This is known as *symmetrical isolation*. Females and males of the same species prefer each other. Females of one species discriminate against the males of alien species.

TABLE 1 : Symmetrical sexual isolation between A and B species (male-choice experiments—one type of males with two types of females—males have choice)

Females	Male	% homogamic matings	%heterogamic matings	Isolation index
A+B	A	90	10	0.80
A+B	B	90	10	0.80

Table 2 shows *asymmetrical mating preference*. With A males, there is strong isolation—B females do not accept A males. On the other hand, both A and B females are equally receptive to B males. So there is random mating with B males.

TABLE 2 : Asymmetrical sexual isolation between A and B species (male-choice experiments)

Females	Male	% homogamic matings	%heterogamic matings	Isolation index
A+B	A	90	10	0.80
A+B	B	50	50	0.00

It is clear from Table 3 that there is *asymmetrical mating preference* but it is just reverse of what has been shown in Table 2. There is strong isolation with B males, but no isolation with A males. A males are readily accepted by both females, but B males are discriminated against by A females.

TABLE 3 : Asymmetrical sexual isolation

Females	Male	% homogamic matings	%heterogamic matings	Isolation index
A+B	A	50	50	0.00
A+B	B	90	10	0.80

The *asymmetrical mode of mating preference* among closely related species as shown in Tables 2 and 3 has been observed. It has been used to predict the direction of evolution among closely related species and two opposing models have been proposed:

(i) In 1976 K. Y. Kaneshiro of Hawaii University, USA, proposed that ancestral females discriminate against derived males but females of

derived-species mate randomly with males of ancestral species. This hypothesis is based on founder principle (evolution via genetic drift) and the mating preference observed among four species of Hawaiian *Drosophila*.

(ii) In 1979 T. K. Watanabe and M. Kawanishi of National Institute of Genetics, Japan, proposed that it is the females of derived species which do not mate with the males of ancestral species. They proposed their hypothesis on the basis of mating preference observed in *D. melanogaster* and *D. virilis* species groups.

Evidence in support of both these hypotheses have been obtained. Thus, the relationship proposed by these scientists can not be generalised for the whole genus *Drosophila*. However, a consistent relationship having predictive value might exist in certain species groups. The relationships might depend upon ecological and evolutionary history of a particular group. The events of speciation may occur in different ways in different cases.

In order to explain how the reproductive isolating mechanisms originate between populations, different hypotheses have been proposed. According to H. J. Muller of Animal Institute of Genetics, Edinburgh, UK, reproductive barriers appear as a side-effect of genetic divergence because populations adapted to different environments acquire genetic differences which lead to reproductive isolation. Th. Dobzhansky of Columbia University, gave more importance to natural selection acting on appropriate genetic variation when the allopatric populations that have an incipient isolation become sympatric. H. L. Carson of University of Hawaii, USA, emphasized the role of genetic drift through founder effect in the origin of reproductive isolation which serves as a barrier to gene flow. Evidences in favour of these theories have been obtained. The development of pre-zygotic isolation as a byproduct of selection or drift is consistent with a widely accepted model of allopatric speciation. In this model, populations become genetically different due to adaptations to different ecological conditions and gradual genetic divergence is accompanied by reproductive in-

compatibility. Isolation has been induced by selection and also by inbreeding in many cases. A large number of investigations on sexual isolation between and within the species of *Drosophila* have indicated the evolutionary origin of sexual isolation. Recently, the phenomenon of sexual isolation has also been studied in *Drosophila* species occurring in India by the present author and his co-workers. Interesting results with respect to the evolutionary significance of this phenomenon have been obtained.

The phenomenon of sexual or ethological isolation is widespread in *Drosophila*. Many cases of incomplete isolation have been reported in various species groups. The pattern and degree of sexual isolation observed among closely related species have often been used to know their phylogenetic relationships. Various stimuli such as visual, auditory and chemical which are

exchanged during courtship between male and female are often used for mate recognition.

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Further readings

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